

Inefficiencies in Local Infrastructure: Evidence from Drinking Water in California

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Infrastructure investment

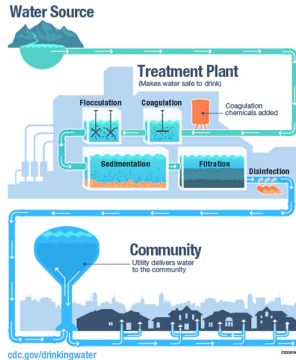
- Growing concerns about infrastructure investment
 - Local governments shape investment in infrastructure
 - ~ half a trillion dollars annually on transportation and water infrastructure
 - Local policymakers may face limited discipline from elections, or weak *electoral accountability*
 - Muted electoral competition at the local level: often uncontested
 - Limited media scrutiny of local policy decisions
- ⇒ Investment may be insufficiently responsive to residents' welfare

Empirical context: Investments in drinking water in California

- Drinking water infrastructure
 - cities and special districts own and operate drinking water systems
 - much of the infrastructure was built in the 1970s and 1980s
 - 10% of Americans exposed to health-based violations [Map](#)
 - despite substantial federal and state subsidies (e.g. \$41.1 billion from since 1997 through Safe Drinking Water Act)
- **This paper**
 1. Quantifies **misallocation** in investment driven by weak electoral accountability.
 2. Shows that weak accountability **undermines the effectiveness** of federal subsidies, by lowering take-up where it matters most

Setting & Data

Provision of drinking water



- 273 cities and 551 special districts
 - ~ 12 districts per county; 2 per schools district
- Infrastructure investment
 - determined by elected officials: city council, county supervisors (dependent districts), district boards (independent districts) [Campaign](#) [Voter response](#)
 - financed by long-term debts (70%) + grants
 - cost is paid by water rate/fee
- Regulated by the Safe Drinking Water Act (1974)
 - 90 pollutants by maximum contaminant levels (MCL)
 - mandates timely public notifications of violations

Data sources, 2003-2022

- **Government investment and water rates** Stat
 - long-term debts (FTR), borrowing terms (CDAIC), federal loans/grants (DWSRF)
 - rates (FTR, eAR, manually collected)
- **Water pollution** Stat
 - contaminants concentration readings (CASWRCB), health standards violations (EPA)
- **Residents preferences for water quality:**
 - house prices (CoreLogic) & house/buyer char. (HMDA + ACS + LEHD), PWS boundaries
- **Local political environment:**
 - elections (CEDA, manually collected), individual turnout (L2)
 - local newspaper presence (3DLNews)

Anecdotal example: Nitrate contamination in San Joaquin valley

- East Orosi totals 99 health based violations since 2003, mainly nitrates from agriculture

“Imagine children knowing that water only comes from a bottle and not from their own tap.”

— Eddie Valero, Tulare County

- Despite documented health risks, in our data Orosi Public Utilities District has
 - no investment during the study period
 - no election records until 2024, since a 2008 recall over alleged misuse of office

Motivating Evidence

Does political environment affect water quality?

- **Outcome:** % of pollutant readings above federal limits (pollutant-jurisdiction-year level)
- **Political environment**
 - competition (% contested elections) and information (local newspaper presence)
- **Challenge:** Political environment correlated with demand and costs
- **Identification: Election timing reform + controls**
 1. e.g., CA Voter Participation Act (2015): move to on-cycle elections
 - higher turnout and more challengers **Mechanism** **DiD**
 - interpreted as a shifter for accountability without directly affecting investment fundamentals
 2. controls for factors affecting costs/benefits of investment (e.g., water source, ..)

Political environment matters

	Average % above MCL			
	(1)	(2)	(3)	(4)
Frequency of contested elections	-0.229*			
	(0.132)			
Any local newspaper within 20 miles		-0.327**		
		(0.145)		
Frequency of on-cycle elections			-0.384**	
			(0.127)	
Holding on-cycle elections				-0.121**
				(0.064)
Median household income (standardized)	-0.228***	-0.233**	-0.233**	-0.0701
	(0.066)	(0.065)	(0.065)	(0.068)
Demographics, system attributes, precipitation stats	Yes	Yes	Yes	Yes
Pollutant-Year FE, County FE	Yes	Yes	Yes	No
Pollutant-Year FE, Pollutant-Jurisdiction FE	No	No	No	Yes
Number of observations	346,196	346,196	346,196	346,196

- Suggest investment is not aligned with welfare $\Rightarrow$ a model to quantify misallocation Cost

Model & Estimation

Model: Accountability and investment

- **Residents:** Value water quality (q_{jt}) vs. cost (r_{jt}) in jurisdiction j with income y_i

$$u(q_{jt}, r_{jt}; y_i) = \omega(y_i) q_{jt} - r_{jt}$$

- **Policymaker:** Chooses investment ($a_{jt} \in \{0, 1\}$) each period, with per-period payoff:

$$\underbrace{\{-\kappa_j + \epsilon_{jt}\}}_{\text{Private cost, limited govt capacity}} \quad a_{jt} + \sum_{i \in \mathcal{I}_{jt}} \lambda_j \pi_{jt}(y_i) \underbrace{\mathbb{E}[u(q_{jt}, r_{jt}, y_i) | a_{jt}]}_{\text{Residents' welfare}}$$

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- **Accountability:** Weight on resident welfare depends on political environment Micro
 - * entry cost of challenger $\Rightarrow$ prob. of contested elections
 - * voter responsiveness (salience of water issues)
 - $\Rightarrow$ Higher accountability (λ_j) $\Rightarrow$ stronger response to welfare gains
 - * relative turnout $\pi_{jt}(y_i)$

Identification and estimation

Estimate dynamic model (Rust, 1987): investment decisions pin down Δ per-period payoff

$$\{-\kappa(x_j) + \epsilon_{jt}\} a_{jt} + \lambda(x_j) \sum_{i \in \mathcal{I}_{jt}} \pi(y_i, x_{jt}) \mathbb{E}[u(q_{jt}, r_{jt}, y_i) | a_{jt}, x_{jt}]$$

- Challenge:** Disentangle residents' preferences (u) and policymakers' preferences (λ, κ)
- Turnout function $\pi(y, x)$ from voting data to weight residents

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- **Estimate residents' net value for investment offline**

Model

Design

Alt.

Results

Robustness

- Housing demand model (Bayer, Ferreira & McMillan, 2007) comparing houses at boundary
- Impact of investment on water quality and rate: event study (Cengiz et al, 2019)

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- **Accountability:** Responsiveness of investment to variation in net value of investment
 - Variation from exogenous factors for pollution and rate (precipitation, subsidy eligibility)

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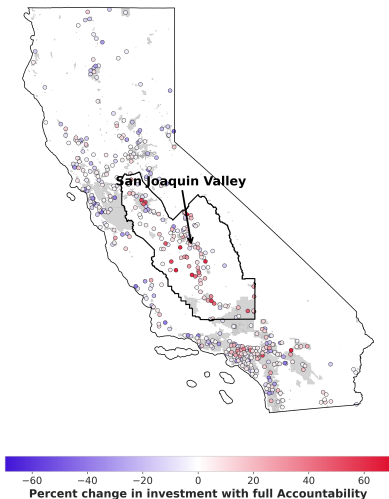
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Finding: Policymakers respond to welfare—more under high accountability Results Fit Direct

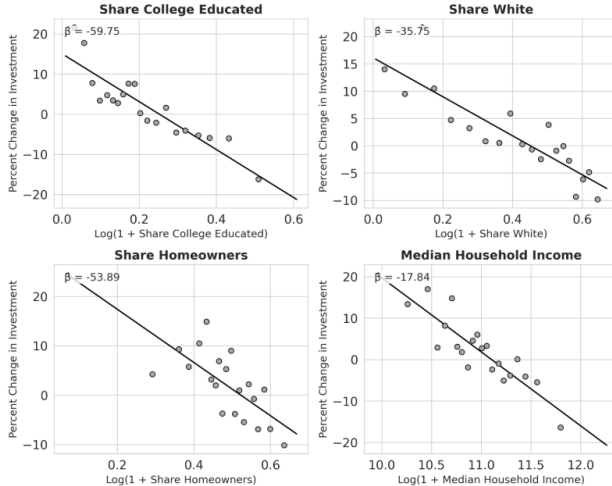
Counterfactuals

Quantifying misallocation from lack of accountability



- What if all elections were contested?
- Changes in investment vary across space
- Improvements concentrated in San Joaquin Valley (agricultural region, polluted water)
 - Includes East Orsi

Progressive consequences of increased accountability

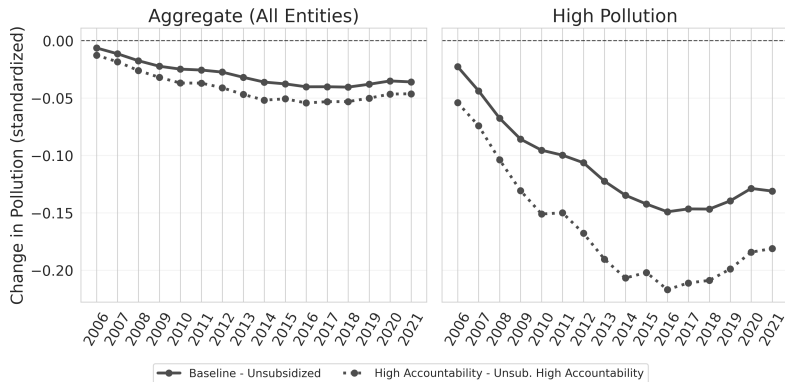


- Low-SES govs exhibit large increases in investment with greater accountability

Role of accountability in the returns to public funding

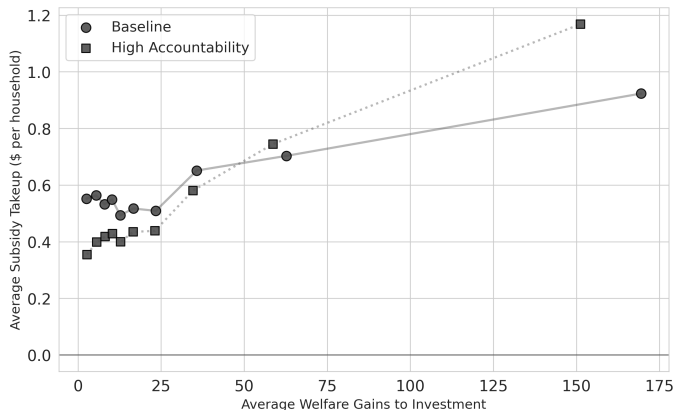
- Substantial public funding for drinking water investment
 - Federal: DWSRF (since 1997), expanded by ARRA (2009) and IIJA (2021)
 - California: general obligation bonds and SB 200 (2019)
 - targeted toward “disadvantaged communities”
- **Does limited electoral accountability reduce the effectiveness of subsidies?**
 - simulate investment under a no-subsidy counterfactual
 - contrast with baseline → effect of subsidies
 - evaluate how these effects change under higher accountability

Accountability improves the targeting of infrastructure policy



- Accountability boosts public funding's impact by 47% in **polluted** systems

Accountability directs subsidies to high-return projects



- Average subsidy take-up amount increases with the welfare gains from investment
- Steeper response under high accountability

Conclusion

- Suggestive evidence:
 - electoral accountability influences drinking water quality
- Empirical framework allows us to:
 - quantify distortions in how residents' preferences are translated into local investment
- Reducing local political distortions is key for effective infrastructure policy
 - accountability improves targeting and efficiency of public funding

Appendix

Roadmap

Measurement of voter preferences

Do voters care?

Impact of switching elections on-cycle

Identification of politicians' payoff parameters

Micro-foundation for the politician's payoff

Additional results

Measuring resident preferences for water quality [Back](#)

1. Hedonic approach using housing transactions [our paper]
2. Avoidance behavior (e.g., water filters or bottled water purchases)
 - Matching purchases to water service areas is imprecise
3. Voting behavior
 - Referenda on water infrastructure bonds
 - * 133 relevant referenda (1995-2022); many bundle multiple purposes
 - Elections for office (city council, county supervisors, water district board)
 - * Enables decomposition of λ
 - * Candidate data not available for water district board elections

⇒ Hedonic approach is the best given data availability

Measuring resident preferences for water quality [Back](#)

- **Interpreting WTP estimates**

- may not capture all residents equally (e.g., homeowners vs. renters)
- capture expectations about future water quality imperfectly (proxied by recent quality)
- based on imperfect information about water quality

- **But they capture revealed preferences of voters**

- Investment responds positively to our measure of net value
- Housing market reacts to water quality shocks
 - * Event study: house prices decline following Tier I EPA violations
- Boundary design: Balance test ✓ + robust to including school characteristics

⇒ WTP provides a meaningful proxy for politically relevant preferences

Robustness to mismeasurement of voter preferences [Back](#)

Concern: Policymakers may observe WTP components we do not

1. **Classical error:** Error orthogonal to Δu
 - attenuation bias in λ (conservative)
2. **Endogenous error** Error correlated with Δu
 - e.g. high-pollution areas have unobservably low demand
 - endogeneity bias

Solution: Control function (work-in-progress)

- Use exogenous shocks to pollution and costs to isolate variation in Δu

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Do voters care about drinking water?

[Back](#)

- Surveys indicate high public concern about drinking water pollution
- Advocacy organizations (e.g., Community Water Center Action Fund)
 - voter education, candidate endorsements, and grassroots campaigns
- Media coverage on water quality conditions and utility performance

 **FONDO DE ACCIÓN**
ACTION FUND

Who We Are Our Work
Support Us Resources

English ^

Donate

Candidate Endorsements

Community Water Center Action Fund (CWCAF) works to elect leaders who view the lack of safe and affordable drinking water in their communities as an environmental justice issue,

 **San Francisco Chronicle**

SALE: ONLY 25¢

Sign In

'Smells like swamp': This Wine Country town is fed up with brown tap water

By **Julie Johnson**, Staff Writer
March 10, 2026

Voters respond to drinking water quality [Back](#)

	Incumbent staying		Incumbent vote share	
	OLS	2SLS	OLS	2SLS
High pollution reading	-0.031** (0.013)	-0.115* (0.061)	-0.033 (0.021)	-0.523*** (0.127)
Year FE	Yes	Yes	Yes	Yes
Observations	4,870	4,870	4,560	4,560
R ²	0.018	-0.006	0.124	-0.350
F-statistics (1st stage)	-	111.90	-	103.65

- Based on election outcome data (not used in the estimation)
- Instruments for past period's pollution: precipitation

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Election timing reshapes local political environments

1. Higher turnout in on-cycle elections (well-established)
2. On-cycle elections are more likely to be contested (our finding)
 - *Candidate-side*: lower entry costs from spillovers from concurrent campaigns
 - *Voter-side*: higher entry returns from more representative electorate
 - *Countervailing force*: stronger incumbency advantage (de Benedictis-Kessner, 2018)→ Net effect is ambiguous a priori

Rather than taking a stand on the mechanism, we treat election timing as an **exogenous shift** in the political environment that affects **accountability**

Election timing and contested elections [Back](#)

	Dependent var: Any contested elections			
	Full sample		Subsample	
	(1)	(2)	(3)	(4)
On-cycle election schedule	0.125*** (0.0326)	0.117*** (0.0330)	0.247*** (0.0474)	0.235*** (0.0506)
Demographic attributes	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes
Jurisdiction FE	Yes	Yes	Yes	Yes
Observations	7,039	6,926	3,230	3,179
R^2	0.562	0.565	0.604	0.606

Notes. The unit of observation is jurisdiction $\times$ two-year period. SE clustered at the jurisdiction level. Columns (1)–(2) use the full sample, while Columns (3)–(4) restrict the data to the post-2014 period and excludes jurisdictions that already had on-cycle election schedules prior to 2018.

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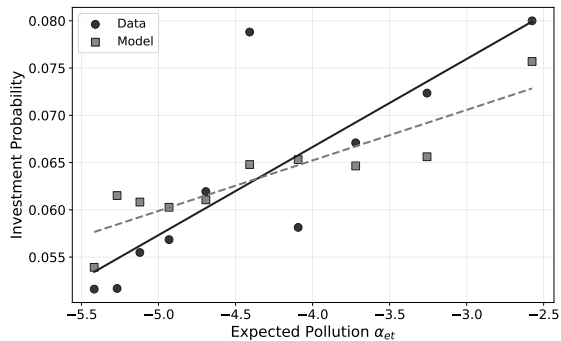
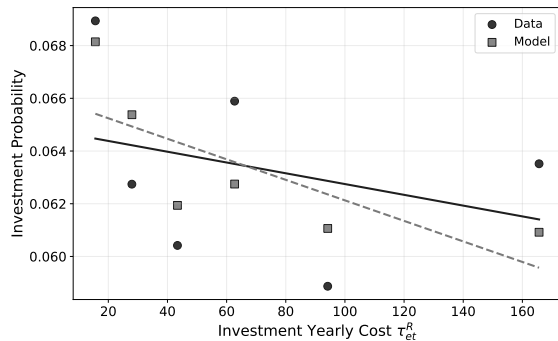
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Additional results

Politicians respond to water investment needs [Back](#)



Responsiveness depends on political environment [Back](#)

	Dependent var.: Any investment		
	(1)	(2)	(3)
Net Utility for Investment (Net WTP)	0.0002*** (5.46e-5)	0.0001** (5.18e-5)	0.0008*** (0.0002)
Net WTP × High Contested	0.0001* (7.29e-5)	0.0001* (7.29e-5)	0.0007** (0.0003)
Observations	8,986	8,986	8,986
R^2	0.0138	0.0190	0.0684
County FE	Yes	Yes	No
Year FE	No	Yes	Yes
Jurisdiction FE	No	No	Yes

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors clustered at the county level.

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Probabilistic voting model [Back](#)

Agents:

- Governing body facing reelection on a staggered cycle $\equiv$ policymaker
- Residents that differ in how they evaluate drinking water cost/quality trade-off

Actions:

- Policymakers make investments with lasting effects on water rates and quality
- Residents vote to retain/replace members based on current outcomes

Timeline:

- Investment decision $\rightarrow$ quality and rate are realized $\rightarrow$ election takes place

Voting and incumbent's expected payoff from elections [Back](#)

- Resident i votes in favor of the policymaker iff

$$u(q_t, r_t, y_i) \geq v_{it}$$

where $v_{it} \sim U[-\frac{1}{2\sigma_v}, \frac{1}{2\sigma_v}]$ independent from water outcomes

- With uncertainty in translating vote share to victory, $\eta_t \sim U[-\frac{1}{2\sigma_\eta}; \frac{1}{2\sigma_\eta}]$,

$$\text{winning prob.} = \frac{1}{2} + \sigma_\eta \sigma_v \int \tilde{\pi}_t(y) u(q_t, r_t, y) dF_{yt}(y)$$

- Incumbent values getting reelected at $\nu +$ elections are contested with prob ρ :

$$\mathbb{E}(\text{payoff from election}) = \underbrace{\rho \nu \sigma_\eta \sigma_v}_\lambda \underbrace{\int \tilde{\pi}_t(y) u(q_t, r_t, y) dF_{yt}(y)}_{\text{turnout-weighted welfare average}}$$

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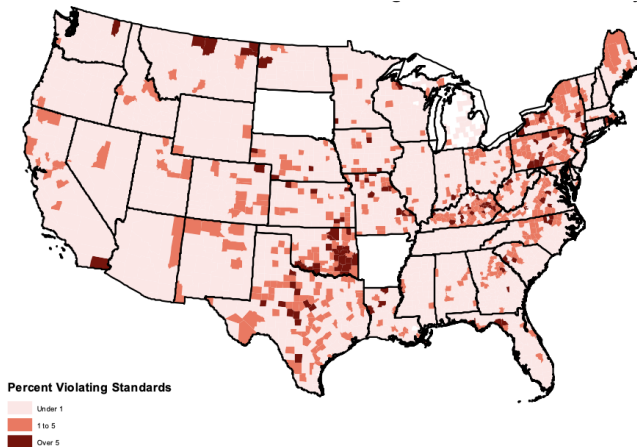
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Additional results

Share of water below health standards, 2009-19

[Back](#)

Source: Kaiser et al (2023), Figure 1-C

Investment frequency and finances [Back](#)

	Cities	Special districts
Annual investment frequency	0.069	0.049
Conditional on any investment		
Amount debt-financed or externally funded		
Mean (in \$ millions)	131.68	48.01
Median (in \$ millions)	13.53	6.96
Per household, Mean (in \$)	3,481	7,971
Per household, Median (in \$)	1,755	2,031
% financed via debts (vs. grants)	66.21	57.92
Number of governments	273	551

Drinking water quality measurement [Back](#)

- Water pollution measured at each treatment facility
- Measured as the sum of
 - (a) an indicator for EPA Tier I and II violations
 - (b) an indicator for each pollutant reading $>$ MCL, excluding disinfection byproductsdivided by (the total number of pollutants + 1)

	Cities	Special Districts
Pollution measure $>$ 0	0.260	0.199
EPA - Tier I	0.005	0.008
EPA - Tier II	0.065	0.084

Accountability increases response to subsidy opportunities [Back](#)

	Dependent var.: Any investment			
	All		Low Accountability	High Accountability
	(1)	(2)	(3)	(4)
E(grant intensity)	0.3909*** (0.1415)	0.4668*** (0.1632)	0.0773 (0.1626)	0.4644*** (0.1637)
E(grant intensity) × Higher accountability		-0.4876** (0.2106)		
Jurisdiction FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	14,320	14,320	1,460	12,860
R ²	0.0602	0.0603	0.0589	0.0609

- Use quasi-exogenous variation in funding: expected grant intensity

Residents' demand for water quality [Back](#)

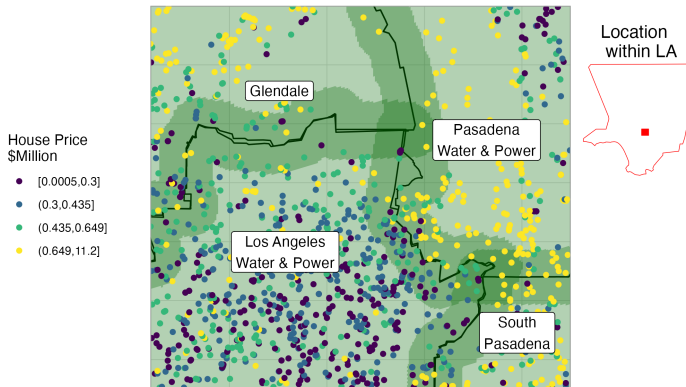
- Individual i 's indirect utility from a house h in period t :

$$U_{iht} = \theta_x \mathbf{x}_{ht} + \theta_q(y_{it})q_{j(h)t} - \theta_p(y_{it})(p_{ht} + r_{j(h)t}) + \theta_d d_{iht} + \psi_{b(h)t} + \xi_{ht} + \varepsilon_{iht}$$

- $q_{j(h)t}$: water quality provided by h 's jurisdiction, $j(h)$
 - $r_{j(h)t}$: annual water rate specific to house h
 - p_{ht} : annual user cost of housing except for r_{ht}
 - d_{iht} : distance to i 's work to house h
- Resident i 's marginal WTP for quality $\omega()$ is given by:

$$\omega(y_{it}) = \frac{\theta_q(y_{it})}{\theta_p(y_{it})}.$$

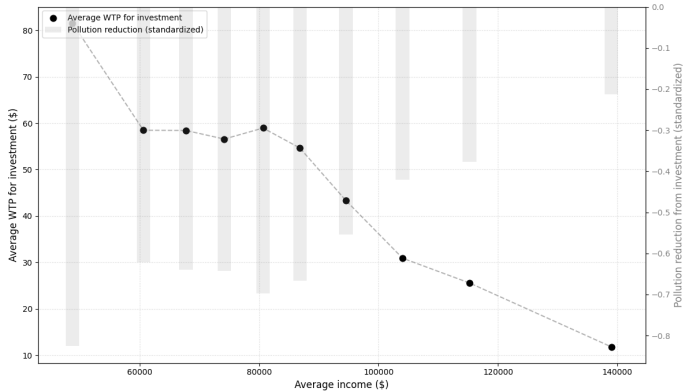
Illustration of boundary design

[Back](#)

Residents care about water quality, *heterogeneously* [Back](#)

	Without Neigh. Pref. (1)	With Neigh. Pref. (2)
First Stage		
Income (log) × Water pollution	0.008 (0.550)	0.025 (0.115)
Income (log) × House price	36.241 (0.044)	32.116 (0.118)
Distance to workplace	-6.792 (0.040)	-6.799 (0.273)
Income (log) × Block-group income		0.480 (0.236)
Second Stage (IV) - Dep. Var: $\hat{\delta}$		
House user cost (price + rate)	-66.187 (1.257)	-59.742 (1.116)
Water pollution	-0.020 (0.007)	-0.023 (0.009)
Fraction of college educated	0.968 (0.047)	0.933 (0.046)
# Bedrooms (log)	0.037 (0.007)	0.038 (0.006)
Other house attributes	Yes	Yes
Border × Year FE, Market FE	Yes	Yes
Observations	870,637	870,637

Willingness to pay for 1 SD increase in water quality [Back](#)



- Mean WTP $\approx$ \$60 per year/household
- Reasonable?
 - family of 4 spend \$250-\$2,700 annually on bottled water
 - average water bill \$1,128

Estimates of policymakers' preferences [Back](#)

	Coefficient	Std. Error
κ: Capacity		
Constant	3.276	(0.371)
Number of skilled employees (log)	-0.157	(0.028)
Small-sized	0.454	(0.039)
λ: Weight for constituent welfare		
Constant	-15.874	(3.160)
Electoral accountability index	9.613	(3.349)
County FE	Yes	
Observations	8,986	
Avg. Neg. log-likelihood	0.230	

Capacity more limited if:

- smaller systems
- fewer skilled employees

Welfare weight increases with electoral accountability